

Kamesh R

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EDUCATION

University College London (UCL)

Master of Science in Machine Learning

London, UK

Aug 2026 – Aug 2027

Sathyabama Institute of Science & Technology

Bachelor of Engineering in Computer Science (AI & ML)

Chennai, India

Sep 2022 – May 2026

PUBLICATIONS & SELECTED ARTICLES

Kamesh R. *Exploring Generalization in NLA's.*

Kamesh R. *Calibrating Activation Vectors using Norm.*

Kamesh R., Sivadhanushya P *Conversational Hallucination Drift: An Episodic Retrieval Framework and Error Taxonomy for Long-Term Memory Evaluation. ICML 2026, Workshop on Failure Modes of Agentic AI*

Kamesh R. *Probing Reasoning Flaws and Safety Hierarchies with Chain-of-Thought Difference Amplification. NeurIPS 2025, LLM-Evals Workshop.*

Kamesh R. *Global PIQA: Evaluating Physical Commonsense Reasoning Across 100+ Languages and Cultures. NeurIPS 2026, Under-review.*

Kamesh R. et al. *Indian Grammatical Tradition-Inspired Universal Semantic Representation Bank (USR Bank 1.0). ACL-IJCNLP 2025, BHASHA Workshop.*

EXPERIENCE

Machine Learning Intern

Glance

Feb 2026 – Present

Bengaluru, India

- Designed an LLM-as-judge evaluation pipeline for catalog recommendations, testing prompting strategies (chain-of-thought, structured rubrics, multi-step verification) to improve judgment reliability and reduce false positives.
- Developed a method to distill human annotator and LLM-judge signals into a cross-encoder, enabling recommendation scoring for zero-usage / cold-start users with no behavioral history.
- Built and optimized catalog retrieval pipelines, improving content discovery across millions of items; achieved a 2x improvement in retrieval metrics by reimplementing SOTA embedding techniques from recent papers.

Independent Interpretability Researcher

Mentored by David Africa (UK AI Security Institute)

Nov 2025 – Feb 2026, Aug 2026 – Present

Remote

- Conducted independent research on mechanistic interpretability, with direct technical guidance from a Research Scientist at the UK AI Safety Institute.
- Developed a framework to permanently embed steering vectors into model weights, enabling persistent safety behaviors without inference-time compute overhead.

Research Fellow

AI Safety Camp (AISC)

Jan 2026 – March 2026

Remote

- Working with **Ihor Kendiukhov** to design and evaluate control protocols (extending Greenblatt et al.) for detecting and bounding misaligned behavior in advanced AI systems.
- Designed hierarchical and parallel control/monitoring topologies to optimize safety-usefulness trade-offs, analogous to multi-layer monitor architectures.
- Ran scaling experiments to test generalization of control and detection guarantees across models of different capability levels.

Machine Learning Contractor

Shipd by Datacurve (YC W24)

Feb 2026 – Present

Remote

- Contributed to high-stakes machine learning challenges for a YC-backed data platform.
- Awarded monetary prizes for top-tier performance in predictive modeling and algorithmic problem-solving tasks.

Research Intern

International Institute of Information Technology (IIIT)

May 2025 – Oct 2025

Hyderabad, India

- Researched Universal Semantic Representations and developed techniques for generating coherent natural language sentences from abstract syntactic-semantic structures.
- Worked on Controlled Image-to-Text Generation systems for scientific images to ensure accurate, context-aware, and domain-specific textual descriptions.

Machine Learning Intern

Dec 2024 – Jan 2025

Co-build.tech

Remote

- Enhanced query performance by 25% by optimizing document embedding strategies for semantic similarity retrieval.
- Automated mapping of legal clause dependencies, improving analysis efficiency by 30%.

GRANTS

Lambda Labs Research Grant — Awarded \$1,000 in initial GPU credits for conducting research on the interpretability and routing dynamics of Mixture-of-Experts (MoE) models during inference.

SERVICE

Reviewer - ICML'26 AI-MATH Workshop

Reviewer - ICML'26 Mechanistic Interpretability Workshop

Reviewer - ICML'26 Failure Modes in Agentic AI Workshop

Reviewer - ICML'26 SCALE Workshop

Reviewer - NeurIPS'25 AI-MATH Workshop

Reviewer - AAAI'26 Workshop on Shaping Responsible Synthetic Data in the Era of Foundation Models

LEADERSHIP & MENTORING

Peer Education Initiative

Fall 2023 – Winter 2024

Sathyabama Institute of Science & Technology

Chennai, India

- Organized and delivered supplementary ML and Deep Learning workshop series for 26 students, covering supervised/unsupervised learning, CNNs, RNNs, and transfer learning.
- Designed curriculum materials and provided hands-on guidance for implementing real-world applications including object detection, NLP tasks, and GANs.
- Delivered guest seminars on neural networks for first-year students, mentoring peers through their initial machine learning projects.

PROJECTS

ManifoldSteer: Geometry-Preserving Activation Steering in LLM Residual Streams

- Mapped the Qwen2.5-3B residual stream to identify ~5-6D semantic submanifolds, establishing an empirical layer-selection criterion for safe steering.
- Designed geometry-preserving steering operators that eliminate the 20x residual stream norm collapse inherent to standard additive CAA baselines.
- Revealed RLHF safety routing occurs 18 layers before peak gradient saliency, proposing an entropy-delta proxy for accurate behavioral targeting.

TALKS & SEMINARS

Presentation – Controlled Image Generation

2025

LTRC, IIT Hyderabad

Hyderabad, India

Talk & Workshop – Prompt Engineering

2025

LTRC, IIT Hyderabad

Hyderabad, India

Lecture – Safety and Alignment of LLMs

2025

Sathyabama Institute of Science & Technology

Chennai, India

Panelist – Impact and Application of GenAI

2024

Google Developer Summit

Chennai, India

TECHNICAL SKILLS

Languages & Tools: Python, C++, SQL, JAX, Linux, WandB, Modal

ML Frameworks: PyTorch, JAX, Transformers (HF), DeepSpeed, vLLM, TRL, Unsloth

Core Libraries: NumPy, SciPy, Pandas, Scikit-learn, Matplotlib, Seaborn, Plotly, Datasets (HF)